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Due: 7/26/2026

STAT 516: Homework 9

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(6 points)** Auto insurance claim amounts (in thousands of dollars) are modeled by a random variable X with probability density function

$$f(x) = xe^{-x}, \qquad x \geq 0.$$

Without a deductible, the insurance company expects to pay all 100 claims it receives.

Suppose the company introduces a deductible of $1,000 (that is, only claims exceeding $1,000 result in a payment).

How many of the 100 claims is the company expected to pay?

2. **(5 points)** If a fair coin is tossed repeatedly, what is the probability that the third head occurs on the nth toss?

3. **(5 points)** As part of the underwriting process for insurance, each prospective policyholder is tested for high blood pressure. Let X represent the number of tests completed when the first person with high blood pressure is found. Suppose

$$E[X] = 12.5.$$

Calculate the probability that the sixth person tested is the first person with high blood pressure.

4. **(7 points)** In a shipment of 20 packages, 7 packages are damaged. The packages are randomly inspected, one at a time and without replacement, until the fourth damaged package is discovered.

Calculate the probability that exactly 12 packages are inspected.

5. **(6 points)** Let X be the number of independent Bernoulli trials performed until one success occurs. Let Y be the number of independent Bernoulli trials performed until five successes occur. A success occurs with probability p, and

$$\text{Var}(X) = \frac{3}{4}.$$

Calculate

$$\text{Var}(Y).$$

6. **(5 points)** Let X have a Poisson distribution with parameter $\lambda = 1$.

Calculate

$$P(X \geq 2 \mid X \leq 4).$$

7. **(5 points)** The number of traffic accidents per week in a small city has a Poisson distribution with mean 3.

   What is the probability of exactly 2 accidents during a two-week period?

8. **(5 points)** Let X, Y, and Z be independent Poisson random variables with

$$E[X] = 3, \qquad E[Y] = 1, \qquad E[Z] = 4.$$

   Calculate

$$P(X + Y + Z \leq 1).$$

9. **(8 points)** Let X have a binomial distribution with parameters n and p. The conditional distribution of Y given $X = x$ is Poisson with mean x.

   Find

$$\mathrm{Var}(Y).$$

10. **(8 points)** The lifetime of a printer costing \$200 is exponentially distributed with mean 2 years. The manufacturer agrees to pay a full refund if the printer fails during the first year following its purchase, a one-half refund if it fails during the second year, and no refund if it fails after the second year.

    Calculate the expected total amount of refunds from the sale of 100 printers.

11. **(9 points)** A piece of equipment is insured against early failure. The time from purchase until failure of the equipment is exponentially distributed with mean 10 years. The insurance pays an amount x if the equipment fails during the first year, and it pays $0.5x$ if the equipment fails during the second or third year. If failure occurs after the first three years, no payment is made.

    Calculate x such that the expected insurance payment is \$1,000.

12. **(5 points)** The time to failure of a component in an electronic device has an exponential distribution with a median of four hours.

    Calculate the probability that the component operates without failure for at least five hours.

13. **(6 points)** An insurance company sells two types of auto insurance policies: Basic and Deluxe. The time until the next Basic policy claim is an exponential random variable with mean two days. The time until the next Deluxe policy claim is an independent exponential random variable with mean three days.

    Calculate the probability that the next claim is a Deluxe policy claim.

14. **(5 points)** A company has two independent electric generators. The time until failure of each generator follows an exponential distribution with mean 10. The company begins using the second generator immediately after the first generator fails.

    Calculate the variance of the total amount of time that the two generators produce electricity.

15. **(7 points)** The joint probability density function of X and Y is

$$f_{X,Y}(x,y) = \begin{cases} 2e^{-(x+2y)}, & x > 0,\ y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find

$$\mathrm{Var}(Y \mid X > 10,\ Y > 3).$$

16. **(4 points)** Let

$$X \sim N(5, 16).$$

Find

$$P(X > 7).$$

17. **(4 points)** Let X be a normal random variable with mean 0 and variance $a > 0$. Compute

$$P(X^2 < a).$$

1. **(6 points)** Auto insurance claim amounts (in thousands of dollars) are modeled by a random variable X with probability density function

$$f(x) = xe^{-x}, \qquad x \geq 0.$$

Without a deductible, the insurance company expects to pay all 100 claims it receives.

Suppose the company introduces a deductible of \$1,000 (that is, only claims exceeding \$1,000 result in a payment). $X>1$

How many of the 100 claims is the company expected to pay? $100P(X>1)$

$P(X>1) = \int_1^\infty xe^{-x}\,dx = \int xe^{-x}\,dx = \left[-(x+1)e^{-x}\right]_1^\infty = 0 - [-2e^{-1}] = 2/e$

$100P(X>1) = 100\left(\frac{2}{e}\right) \approx 73.58 \rightarrow$ 74 claims

2. **(5 points)** If a fair coin is tossed repeatedly, what is the probability that the third head occurs on the nth toss?

Place 2 heads among the first n-1 tosses: $\binom{n-1}{2}$

Fair coin: $\left(\frac{1}{2}\right)^n$

$P(3^{rd}$ head occurs on toss $n) = \binom{n-1}{2}\left(\frac{1}{2}\right)^n$

$P(X=n) = \binom{n-1}{2}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^{n-3}$

3. **(5 points)** As part of the underwriting process for insurance, each prospective policyholder is tested for high blood pressure. Let X represent the number of tests completed when the first person with high blood pressure is found. Suppose

Geometric RV: Counting the # of trials until the first success

$E[X]=1/p$

$$E[X] = 12.5.$$

$P(X=6)$

Calculate the probability that the sixth person tested is the first person with high blood pressure.

$E[X] = 1/p = 12.5$

$p = 0.08$

$1-p = 0.92$

$P(X=6) = (1-p)^5 p = (0.92)^5(0.08)$

$P(X=6) \approx 0.0527$, 5.27%

4. **(7 points)** In a shipment of 20 packages, 7 packages are damaged. The packages are randomly inspected, one at a time and without replacement, until the fourth damaged package is discovered.

   Calculate the probability that exactly 12 packages are inspected.

first 3 damaged packages among 11 inspections

Place the remaining 3 damaged packages among positions 13-20.

$$P(N=12) = \frac{\binom{11}{3}\binom{8}{3}}{\binom{20}{7}}$$

Place all 7 damaged packages among 20 positions.

$$= \frac{165 \cdot 56}{77520}$$

$P(N=12) = 77/646 \approx 0.1192$, 11.92%

5. **(6 points)** Let X be the number of independent Bernoulli trials performed until one success occurs. Let Y be the number of independent Bernoulli trials performed until five successes occur. A success occurs with probability p, and

$X \sim \text{Geometric}(p)$

$$\text{Var}(X) = \frac{3}{4}. = \frac{1-p}{p^2}$$

Calculate

$$\text{Var}(Y).$$

$\frac{1-p}{p^2} = \frac{3}{4}$

$4(1-p) = 3p^2$

$3p^2 + 4p - 4 = 0$

$(3p-2)(p+2) = 0$

$p = 2/3$, $0<p<1$

$Y \sim \text{Neg. Binomial}(r=5, p)$

$\text{Var}(Y) = \frac{r(1-p)}{p^2} = 5\left(\frac{1-p}{p^2}\right) = 5\,\text{Var}(X) = 5\left(\frac{3}{4}\right)$

$\text{Var}(Y) = 15/4$

6. **(5 points)** Let X have a Poisson distribution with parameter $\lambda = 1$. $X \sim \text{Poisson}(1)$

Calculate
$$P(X \geq 2 \mid X \leq 4).$$

$$P(X \geq 2 \mid X \leq 4) = \frac{P(X > 2, X \leq 4)}{P(X \leq 4)} = \frac{P(2 \leq X \leq 4)\ ①}{P(X \leq 4)\ ②}$$

$$P(X = x) = e^{-\lambda}\frac{\lambda^x}{x!} \xrightarrow[\lambda = 1]{} \frac{e^{-1}}{x!}$$

① $P(2 \leq X \leq 4) = P(X=2) + P(X=3) + P(X=4) = e^{-1}\left(\frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!}\right) = e^{-1}\left(\frac{1}{2} + \frac{1}{6} + \frac{1}{24}\right) = e^{-1}\left(\frac{17}{24}\right)$

② $P(X \leq 4) = \sum_{x=0}^{4}\left(\frac{e^{-1}}{x!}\right) = e^{-1}\left(1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24}\right) = e^{-1}\left(\frac{65}{24}\right)$

$$P(X \geq 2 \mid X \leq 4) = \frac{P(2 \leq X \leq 4)}{P(X \leq 4)} = \frac{\cancel{e^{-1}}\left(\frac{17}{\cancel{24}}\right)}{\cancel{e^{-1}}\left(\frac{65}{\cancel{24}}\right)}$$

$$P(X \geq 2 \mid X \leq 4) = \frac{17}{65} \approx 0.2615$$

7. **(5 points)** The number of traffic accidents per week in a small city has a Poisson distribution with mean 3.

What is the probability of exactly 2 accidents during a two-week period?

Mean $\lambda = 2(3) = 6$

$X \sim \text{Poisson}(6)$

Poisson PMF $= P(X = x) = e^{-\lambda}\left(\frac{\lambda^x}{x!}\right)$

$$P(X=2) = e^{-6}\left(\frac{6^2}{2!}\right) = e^{-6}\left(\frac{36}{2}\right) = 18e^{-6}$$

$$P(X=2) \approx 0.0446$$

8. **(5 points)** Let X, Y, and Z be independent Poisson random variables with

$$E[X] = 3, \qquad E[Y] = 1, \qquad E[Z] = 4.$$

$X \sim \text{Poisson}(3)$ $X \sim \text{Poisson}(1)$ $X \sim \text{Poisson}(4)$

Calculate

$$\underbrace{P(X+Y+Z}_{W} \leq 1). \sim \text{Poisson}(8)$$

$P(W=w) = e^{-8}\frac{8^w}{w!}$

$P(W\leq 1) = P(W=0) + P(W=1) = e^{-8}\frac{8^0}{0!} + e^{-8}\frac{8^1}{1!} = e^{-8}(1+8) = 9e^{-8}$

$P(X+Y+Z \leq 1) = 9e^{-8} \approx 0.00302$

9. **(8 points)** Let X have a binomial distribution with parameters n and p. The conditional distribution of Y given $X = x$ is Poisson with mean x.

$X \sim Bin(n,p)$

Find

$$\text{Var}(Y).$$

$Y|X=x \sim \text{Poisson}(x)$

$Var(Y) = E[Var(Y|X)] + Var(E[Y|X])$

Poisson RV with mean x, $E[Y|X=x] = x \longrightarrow E[Y|X] = X$

$Var(Y|X=x) = x \longrightarrow Var(Y|X) = X$

$Var(Y) = E[X] + Var(X)$

$X \sim Bin(n,p)$: $E[X] = np$

$Var(X) = np(1-p)$

$Var(Y) = np + np(1-p) = np[1+(1-p)] = np(2-p)$

$Var(Y) = np(2-p)$

$\lambda = 1/2$

10. **(8 points)** The lifetime of a printer costing \$200 is exponentially distributed with mean 2 years. The manufacturer agrees to pay a full refund if the printer fails during the first year following its purchase, a one-half refund if it fails during the second year, and no refund if it fails after the second year. Exponential

Calculate the expected total amount of refunds from the sale of 100 printers.

Exponential RV: $P(T>t) = e^{-\lambda t} = e^{-t/2}$

$$R = \begin{cases} 200, & T \le 1 \\ 100, & 1 < T < 2 \\ 0, & T > 2 \end{cases}$$

For one printer: $E[R] = 200\,\underbrace{P(T\le 1)}_{1} + 100\,\underbrace{P(1<T\le 2)}_{2}$

1. $P(T\le 1): 1 - P(T>1) = 1 - e^{-1/2}$
2. $P(1<T<2): P(T>1) - P(T>2) = e^{-1/2} - e^{-1}$

$E[R] = 200(1-e^{-1/2}) + 100(e^{-1/2} - e^{-1}) = 200 - 100e^{-1/2} - 100e^{-1}$

For 100 Printers: $100(200 - 100e^{-1/2} - 100e^{-1}) \approx 10255.90$ \$

$\lambda = 1/10$

11. **(9 points)** A piece of equipment is insured against early failure. The time from purchase until failure of the equipment is exponentially distributed with mean 10 years. The insurance pays an amount x if the equipment fails during the first year, and it pays $0.5x$ if the equipment fails during the second or third year. If failure occurs after the first three years, no payment is made.

Calculate x such that the expected insurance payment is \$1,000.

$P(T>t) = e^{-t/10}$

$$R = \begin{cases} X, & 0 < T \le 1 \\ 0.5X, & 1 < T \le 3 \\ 0, & T > 3 \end{cases}$$

$E[R] = 1000$

$P(T\le 1) = 1 - e^{-0.1}$

$P(1<T\le 3) = P(T>1) - P(T>3) = e^{-0.1} - e^{-0.3}$

$E[R] = XP(T\le 1) + 0.5XP(1<T\le 3) = 1000$

$X(1-e^{-0.1}) + 0.5X(e^{-0.1} - e^{-0.3}) = 1000$

$X\left[1 - e^{-0.1} + \frac{1}{2}(e^{-0.1} - e^{-0.3})\right] = 1000$

$$X = \frac{1000}{1 - 0.0904 + \frac{1}{2}(0.0904 - 0.7408)}$$

$X \approx 5644.23$ \$

12. **(5 points)** The time to failure of a component in an electronic device has an exponential distribution with a median of four hours. Exponential w/ rate λ

Calculate the probability that the component operates without failure for at least five hours.

$P(T>t) = e^{-\lambda t} = -e^{4\lambda} = \frac{1}{2}$ $\quad P(T \le \text{median}) = \frac{1}{2}$ $\quad P(T \ge 5)$

$-4\lambda = \ln\left(\frac{1}{2}\right) = -\ln 2$ $\quad P(T > \text{median}) = \frac{1}{2}$

$\lambda = \frac{\ln 2}{4}$

$P(T \ge 5) = e^{-5\lambda} = e^{-5\ln 2/4} = 2^{-5/4}$

$e^{\ln a} = a$

$P(T \ge 5) \approx 0.4204$

13. **(6 points)** An insurance company sells two types of auto insurance policies: Basic and Deluxe. The time until the next Basic policy claim is an exponential random variable with mean two days. The time until the next Deluxe policy claim is an independent exponential random variable with mean three days.

TB

TD

Exponential with rate(s) $\lambda_B = \frac{1}{2}$

$\lambda_D = \frac{1}{3}$

Calculate the probability that the next claim is a Deluxe policy claim.

$P(T_D < T_B)$

$P(T_D < T_B) = \frac{\lambda_D}{\lambda_B + \lambda_D} = \frac{1/3}{1/2 + 1/3}$

$P(T_D < T_B) = 2/5 = 0.4$

14. **(5 points)** A company has two independent electric generators. The time until failure of each generator follows an exponential distribution with mean 10. The company begins using the second generator immediately after the first generator fails. Exponential

$\lambda = 1/10$

Calculate the variance of the total amount of time that the two generators produce electricity.

$T = X_1 + X_2$

$E[X_i] = 10$

$Var(X_i) = 1/\lambda^2 = 1/(1/10)^2 = 10^2 = 100$

$Var(T) = Var(X_1) + Var(X_2) = 100 + 100$

$Var(T) = 200$

15. **(7 points)** The joint probability density function of X and Y is

$$f_{X,Y}(x,y) = \begin{cases} 2e^{-(x+2y)}, & x > 0,\ y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

$e^{-x} \cdot 2e^{-2y} = f_X(x) f_Y(Y)$

$X \sim Exp(1)$, $Y \sim Exp(2)$
independent

Find

$$\mathrm{Var}(Y \mid \cancel{X > 10},\ Y > 3).$$

no addition info about Y

$\mathrm{Var}(Y \mid Y>3)$

Exponential RV: $P(Y>s+t \mid Y>s) = P(Y>t)$

once Y exceeded 3, additional waiting time beyond 3 has the original exponential dist.

$Y-3 \mid Y>3 \sim Exp(2)$

$Y \mid Y>3 \overset{d}{=} 3+Z$ where $Z \sim Exp(2)$

$\mathrm{Var}(Z + \text{constant}) = \mathrm{Var}(Z)$

$\mathrm{Var}(Y \mid Y>3) = \mathrm{Var}(3+Z) = \mathrm{Var}(Z)$

$\mathrm{Var}(Z) = \frac{1}{\lambda^2} = \frac{1}{2^2}$

$\mathrm{Var}(Z) = \frac{1}{4} = \mathrm{Var}(Y \mid X>10, Y>3)$

16. **(4 points)** Let

$u=5,\ \sigma^2=16,\ \sigma=4$

$$X \sim N(5, 16).$$

Find

$$P(X > 7).$$

Standardize: $Z = \frac{X-u}{\sigma}$

$P(X>7) = P\left(\frac{X-5}{4} > \frac{7-5}{4}\right) = P(Z>0.5) = 1 - \underbrace{\Phi(0.5)}_{0.6915\ldots} = 0.3085$

under standard normal CDF

$P(X>7) = 0.3085$

$u=0$ $\sigma^2=a$ $\sigma=\sqrt{a}$

17. **(4 points)** Let X be a normal random variable with mean 0 and variance $a > 0$. Compute

$$P(X^2 < a). = P(-\sqrt{a} < X < \sqrt{a})$$

Standardize: $Z = \frac{X}{\sqrt{a}} \sim N(0,1)$

$P(X^2 < a) = P(-1 < Z < 1) = \Phi(1) - \Phi(-1) = 2\underbrace{\Phi(1)}_{0.8413\ldots} - 1 = 2(0.8413) - 1$

$P(X^2 < a) \approx 0.6827$